

MTH603

Assignment 1

Solution BY VU360

Question 1 Solution By Mudasar Qureshi Vu360:

Regula Falsi Method

The Regula Falsi method, also known as the false position method, is a numerical method used to find the roots of a real – valued function. It is a simple and efficient method for finding the root of a function.

Given Equation

$$xe^x - 3 = 0$$

Interval

$$[1, 1.5]$$

Regula Falsi Method Steps

Choose two initial guesses x_0 and x_1 within the interval $[1, 1.5]$. Let's choose $x_0 = 1$ and $x_1 = 1.5$.

Evaluate the function at these points: $f(x_0) = 1e^1 - 3 = -1.6321$ and $f(x_1) = 1.5e^{1.5} - 3 = 1.3817$.

$$\text{Calculate the secant line slope: } m = \frac{(f(x_1) - f(x_0))}{(x_1 - x_0)} = \frac{(1.3817 + 1.6321)}{(1.5 - 1)} = 3.0138.$$

$$\text{Calculate the new estimate: } x_2 = x_1 - \frac{f(x_1)}{m} = 1.5 - \frac{1.3817}{3.0138} = 1.2414.$$

Repeat steps 2 – 4 until convergence or desired accuracy.

Vu360° Official: <https://chat.whatsapp.com/F5zhJsJtf5yL4RjDODz1fL>

Vu360° 1 : <https://chat.whatsapp.com/HJYBCFqTTpX3QYAHhOLSa2>

Vu360° Discussion ✨

<https://chat.whatsapp.com/IlukS0LK2Uh15bOeBN2IJw>

Iterations:

i	x_i	$f(x_i)$	m	$x_{(i+1)}$
0	1.0000	-1.6321	-	-
1	1.5000	1.3817	3.0138	1.2414
2	1.2414	0.0536	3.0055	1.2429
3	1.2429	0.0004	3.0055	1.2430

The root of the equation $xe^x - 3 = 0$ in the interval $[1, 1.5]$ is approximately $x = 1.2430$.

Note: The Regula Falsi method converges quadratically, meaning the number of correct digits in the estimate roughly doubles with each iteration. The actual root is $x \approx 1.24298$.

Question2 Solution By Mudasar Qureshi Vu360:*Newton Raphson Method*

The Newton Raphson method is a numerical method used to find the roots of a real – valued function. It is a powerful and efficient method for finding the root of a function.

Given Equation

$$\sin(x) - 1 - x^3 = 0$$

Interval

$$[-2, -1]$$

Initial Value

$$x_0 = -1.1$$

Newton Raphson Method Steps

Evaluate the function at the initial value: $f(x_0) = \sin(-1.1) - 1 - (-1.1)^3 = -2.2444$.

Vu360° Official: <https://chat.whatsapp.com/F5zhJsJtf5yL4RjDODz1fL>

Vu360° 1 : <https://chat.whatsapp.com/HJYBCFqTTpX3QYAHhOLSa2>

Vu360° Discussion ✨

<https://chat.whatsapp.com/IlukS0LK2Uh15bOeBN2IJw>

Evaluate the derivative of the function at the initial value: $f'(x_0) = \cos(-1.1) - 3 * (-1.1)^2 = 2.3161$.

Calculate the new estimate: $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = -1.1 - \frac{(-2.2444)}{2.3161} = -1.0535$.

Repeat steps 1 – 3 until convergence or desired accuracy.

Iterations:

i	x_i	$f(x_i)$	$f'(x_i)$	$x_{(i+1)}$
0	-1.1000	-2.2444	2.3161	-1.0535
1	-1.0535	-0.1429	2.1039	-1.0463
2	-1.0463	-0.0063	2.0713	-1.0462

Root (after three iterations)

The root of the equation $\sin(x) - 1 - x^3 = 0$ in the interval $[-2, -1]$ is approximately $x = -1.0462$.

Note: The Newton Raphson method converges quadratically, meaning the number of correct digits in the estimate roughly doubles with each iteration.

Vu360° Official: <https://chat.whatsapp.com/F5zhJsJtf5yL4RjDODz1fL>

Vu360° 1 : <https://chat.whatsapp.com/HJYBCFqTTpX3QYAHhOLSa2>

Vu360° Discussion ✨

<https://chat.whatsapp.com/IlukS0LK2Uh15bOeBN2IJw>